

PROJECT REPORT

Diabetes Risk Prediction using Machine Learning

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1. Introduction

Diabetes is a chronic medical condition that occurs when the body cannot effectively regulate blood sugar levels. Given the rising global health trends, early detection is vital. This project aims to utilize Machine Learning algorithms to predict the risk of diabetes based on clinical health data such as Glucose levels, BMI, and Age.

2. Problem Statement

Manual medical diagnosis can sometimes be time-consuming or prone to human error in the early stages. By using a data-driven approach, we can identify subtle patterns in medical records that indicate future health risks, thereby assisting in preventive healthcare and timely intervention.

3. Objective

The primary goal is to build a robust predictive model using the **Random Forest Classifier** to determine whether a patient is at risk of diabetes based on specific diagnostic measurements.

4. System Architecture & Methodology

The project follows a standard Machine Learning pipeline:

1. **Data Acquisition:** Using the Pima Indians Diabetes Dataset.
2. **Data Preprocessing:** Identifying and handling invalid "zero" values in physiological parameters.
3. **Exploratory Data Analysis (EDA):** Utilizing Heatmaps and Statistical Summaries to understand feature correlations.
4. **Model Development:** Splitting data into training/testing sets and applying the Random Forest algorithm.
5. **Performance Evaluation:** Measuring success via Accuracy Scores and Confusion Matrices.

5. Implementation Highlights

Data Cleaning & Imputation

In this dataset, columns like Glucose, BMI, and BloodPressure contained zero values, which are biologically impossible for a living patient. These were treated as missing data:

Python

```
# Replacing 0 with NaN and filling with mean values
```

```
cols = ['Glucose','BloodPressure','SkinThickness','Insulin','BMI']
```

```
df[cols] = df[cols].replace(0, np.nan)
```

```
df.fillna(df.mean(), inplace=True)
```

Algorithm Selection

The **Random Forest Classifier** was chosen for its ability to handle non-linear relationships and its resistance to overfitting. By aggregating the results of multiple decision trees, the model provides a more stable and accurate prediction.

6. Results and Analysis

- **Correlation Insights:** The Heatmap revealed that Glucose and BMI have the strongest positive correlation with the diabetes Outcome.
- **Model Performance:** The model achieved high accuracy on the test set, demonstrating its reliability in classification.
- **Feature Importance:** Using the model's internal ranking, we visualized that Glucose levels are the most significant predictor for this specific dataset.

7. Conclusion

This project demonstrates the efficiency of Machine Learning in the healthcare sector. By automating the risk assessment process, healthcare providers can prioritize high-risk patients and encourage lifestyle changes or medical treatments much earlier than traditional methods might allow.

8. Future Enhancements

- Integrating a **Streamlit** or **Flask** web interface for real-world usability.
- Applying **Hyperparameter Tuning** (GridSearchCV) to further optimize model accuracy.
- Expanding the dataset to include more diverse demographics for better generalization.